

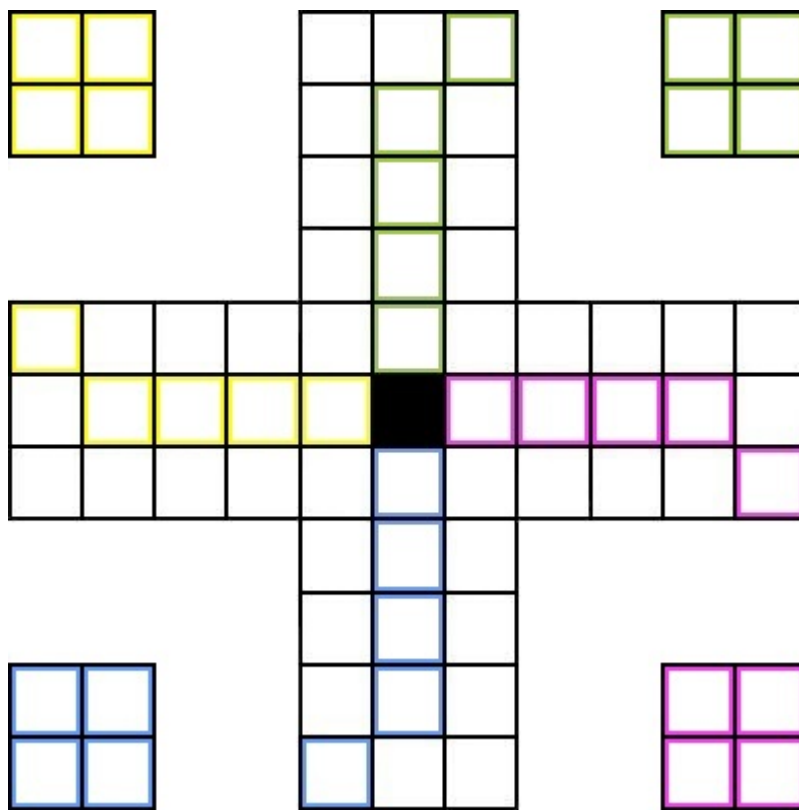
Board Game Video Detection Project

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1. Board Game - Man, Don't Get Angry (w/ coins)

1.1 Description

It is a board game for 2-4 players. Each player has 4 pawns of their color that start in their home base. Players take turns rolling a dice to move their pawns around the board, trying to get all four pawns from the start area to their finish zone. A roll of 6 is required to move a pawn onto the board and grants an extra turn. If a pawn lands on a space occupied by an opponent's pawn, the opponent's pawn is captured and sent back to their home base. In our extended version, capturing an opponent's pawn awards the capturing player a coin token.



1.2 Items

- Game board - Cross-shaped board with 4 colored corners (home bases), 4 finish zones
- Pawns – 16 total (4 per player in 4 different colors)
- Die – Standard 6 sided die
- Coin tokens – Awarded when a player captures an opponent's pawn

1.3 Events

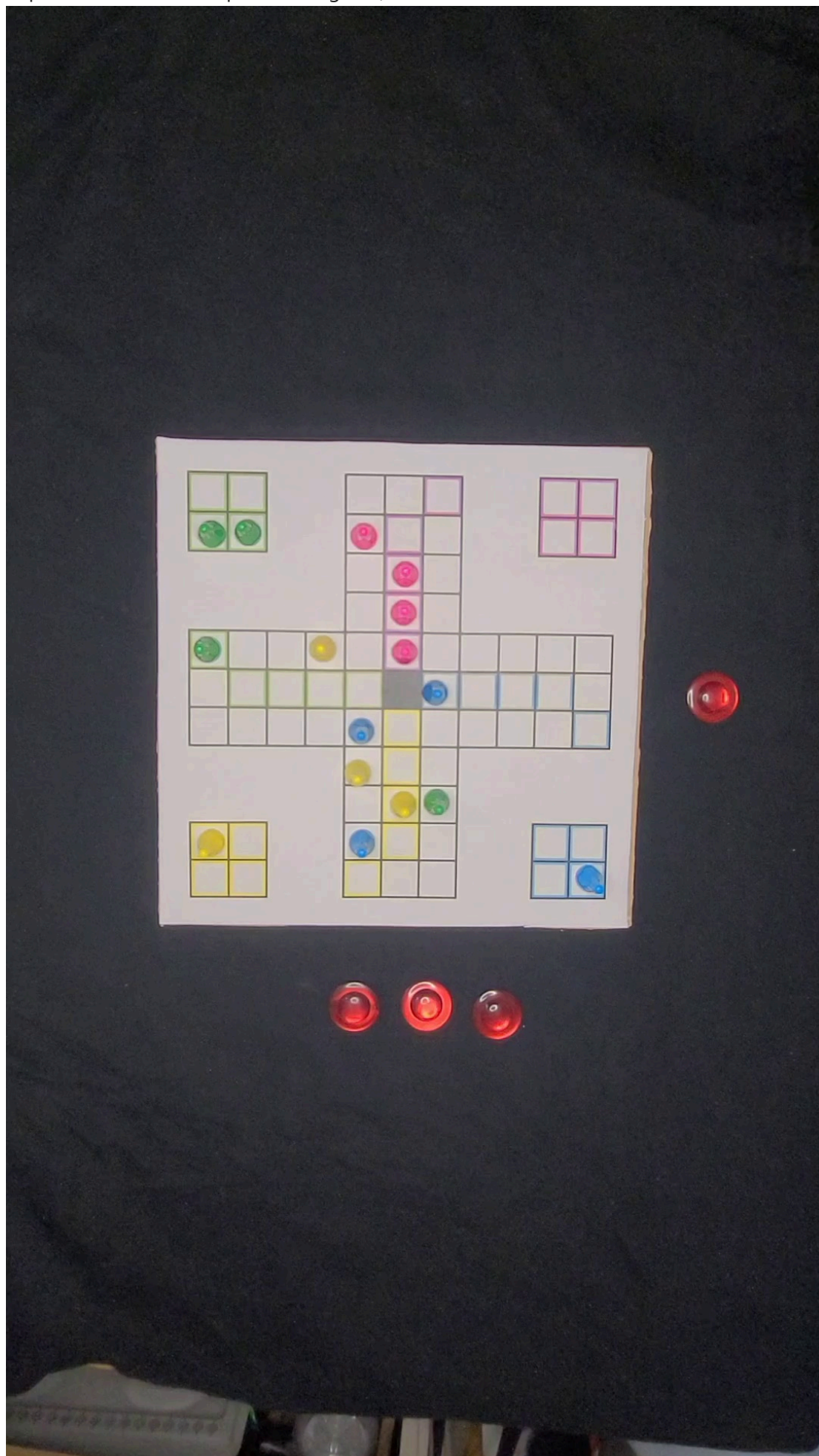
- Dice roll - Reading the value on the dice
- Pawn entering the board - and bonus turn (rolling 6)
- Moving a pawn
- Capture opponent/Coin collection - A pawn lands on an opponent's pawn, sending it back to home base, capturing player receives a coin token
- Pawn entering finish zone/No more moves - A pawn moves into the finish zone and can no longer be moved
- Game start – All pawns are positioned in their respective home bases
- Player wins/Game end – A player successfully moves all 4 of their pawns into the finish zone

2. Dataset

2.1 Easy

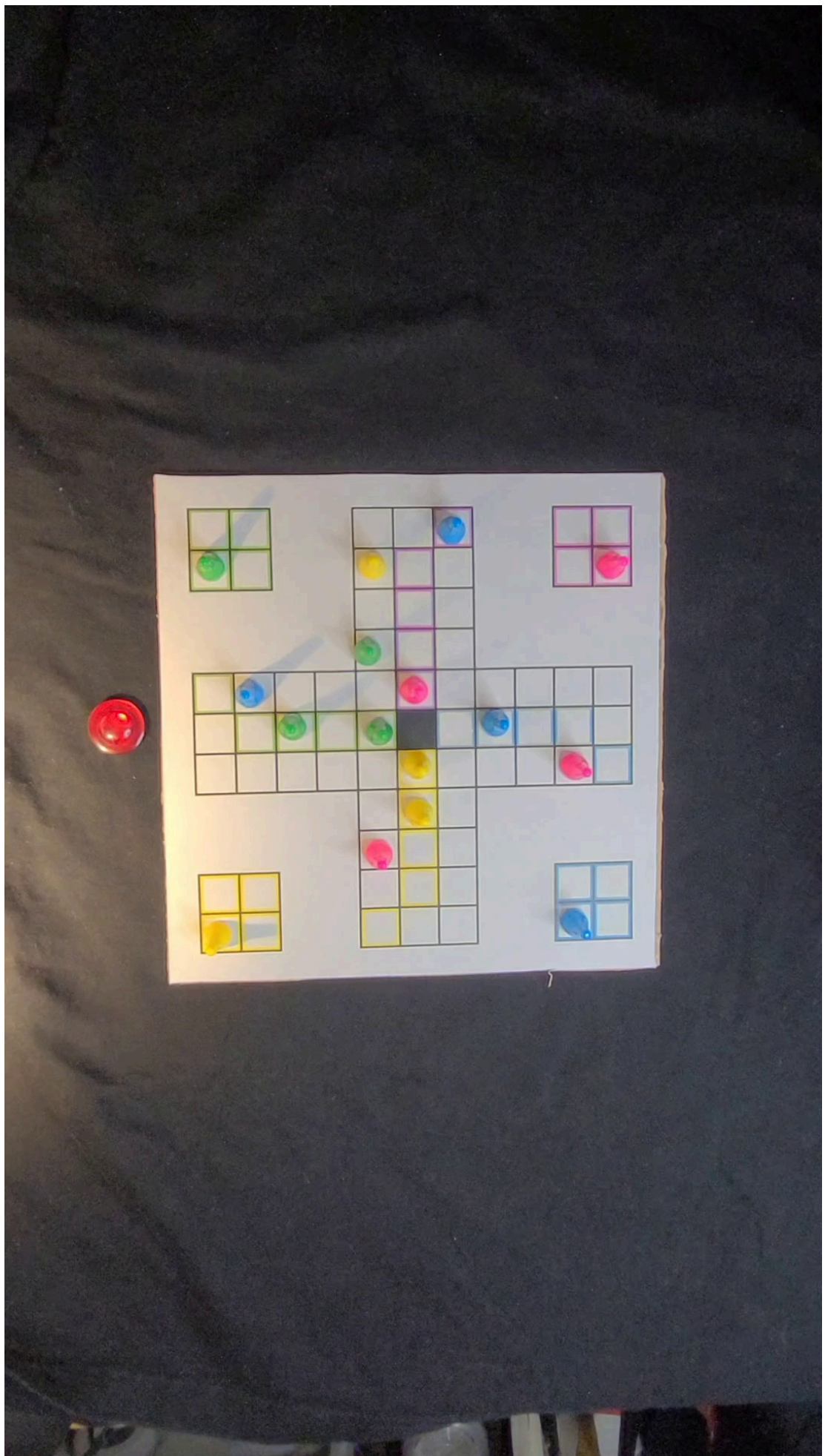
Videos were recorded from a top-down view with lighting conditions as good as possible. Every part of the game was clearly visible. Players hand obstructed the pieces only when it was needed (due to the size and shape of the board many times it was

impossible to not cover part of the game).



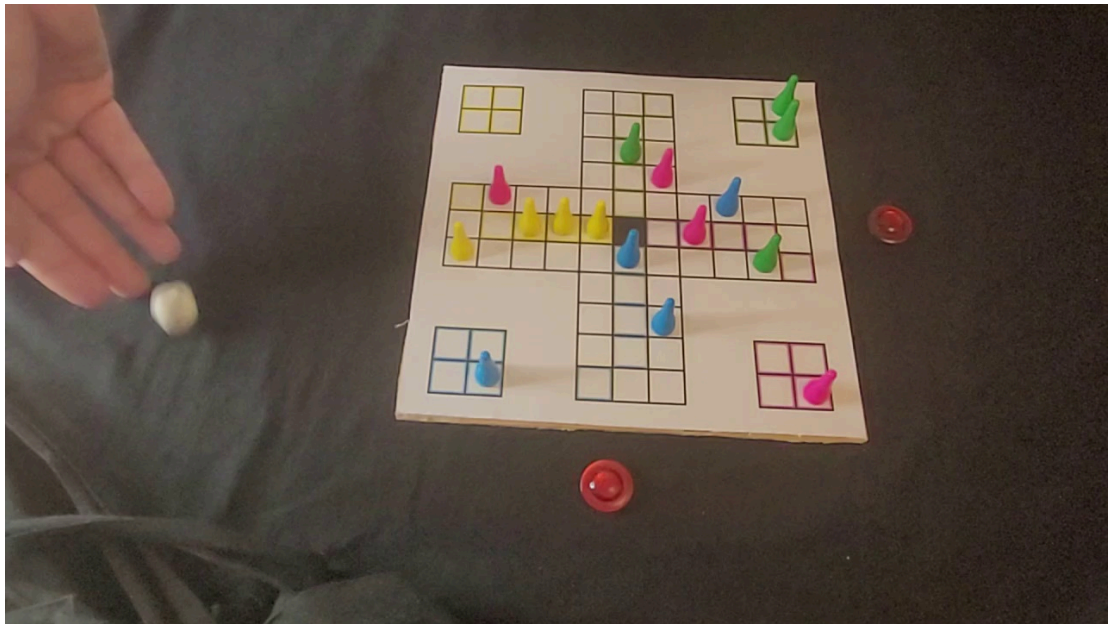
2.2 Medium

Same as for easy dataset, except the lighting conditions were not as good and created shadows, light reflections, darker board.



2.3 Difficult

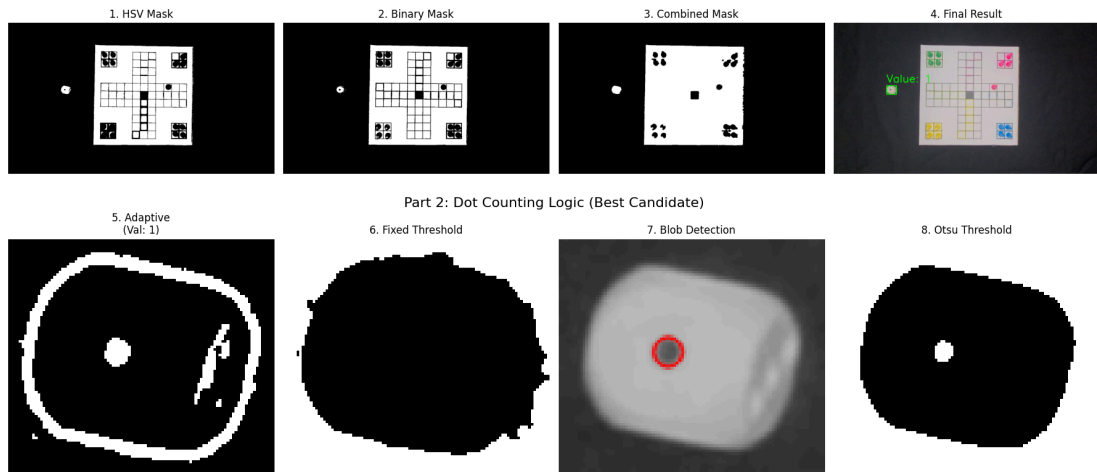
Same as for medium, except the angle of the camera had several degrees with slight shaking sometimes and the hand obstructed the game pieces more often.



3. Methods used

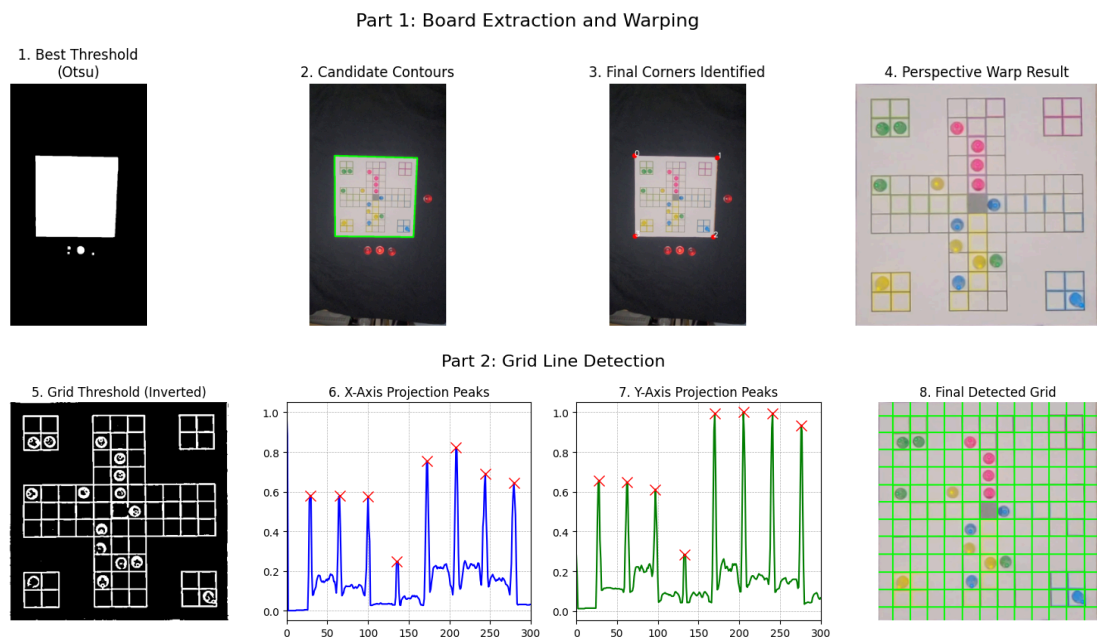
3.1 Dice detection

The detection function isolates the dice by converting the input frame to the HSV color space and generating a binary mask based on brightness and saturation thresholds. It refines this mask using morphological closing and opening operations, which fill internal holes and remove background noise to ensure a solid shape. The function then filters potential candidates by analyzing geometric properties, rejecting contours that do not meet specific area and circularity criteria typical of a die. Finally, it determines the dice value using adaptive thresholding, which dynamically separates black pips from the white surface by calculating local thresholds, ensuring accuracy despite uneven lighting. There are multiple methods used to detect the pips and they vote on what the value should be.



3.2 Board detection

To isolate the board from the background, we began by converting the input frame to grayscale and applying a Gaussian blur to reduce image noise. Later rather than relying on a single binarization parameter, we generated distinct binary candidate maps utilizing three diverse techniques: Otsu's method for global threshold optimization, a fixed empirical threshold, and adaptive Gaussian thresholding. Each candidate map underwent morphological opening and closing. We then performed contour analysis for all candidates, evaluating the geometric properties of the detected shapes to identify the best quadrilateral that maximized the board area. The vertices of this optimal candidate were subsequently mapped to a rectified coordinate system to compute the grid matrix for the perspective warp using the peaks to find the lines.



3.3 Pawns detection

To detect and track the game pieces, we used the grid to extract a region of interest (ROI) for each logical tile (to find each colors home, starting and finish tiles). Pixel analysis

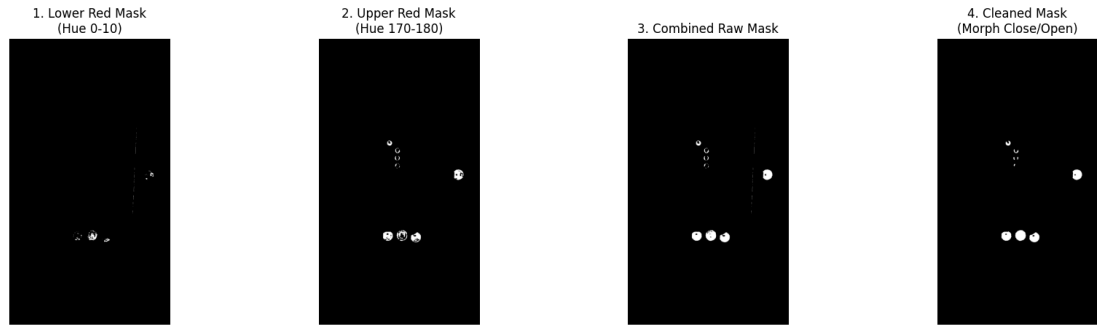
Figure 1 illustrates the pipeline of the proposed method, consisting of four main stages:

- 1. Grid Extraction:** A 10x10 grid is shown with various colored circles (green, pink, yellow, blue) representing different objects or features.
- 2. Color Extraction (Raw):** The grid is converted into a raw color representation, where each cell is colored based on its content (e.g., green for green circles, pink for pink circles).
- 3. Square Scoring (ROI):** The raw color representation is processed to generate a square scoring map (ROI), where each cell is assigned a score (e.g., 10, 9, 8, 7, 6, 5, 4, 3, 2, 1).
- 4. Final Tracker Output:** The final tracker output is shown, where the grid is colored based on the square scoring map, and the final scores are displayed for each cell (e.g., 10, 9, 8, 7, 6, 5, 4, 3, 2, 1).

The detection function isolates tokens by converting the input frame to the HSV color space and generating a binary mask from two separate hue ranges to capture the red color spectrum's wrap-around nature. It improves this mask using morphological closing and opening operations, to eliminate background noise and produce clean, continuous contours. Then it filters these contours by analyzing their geometric properties, rejecting candidates that do not meet area and circularity thresholds of those tokens. Finally, it employs Euclidean distance tracking to associate new detections with existing token identities, ensuring consistent tracking across consecutive frames despite object movement.

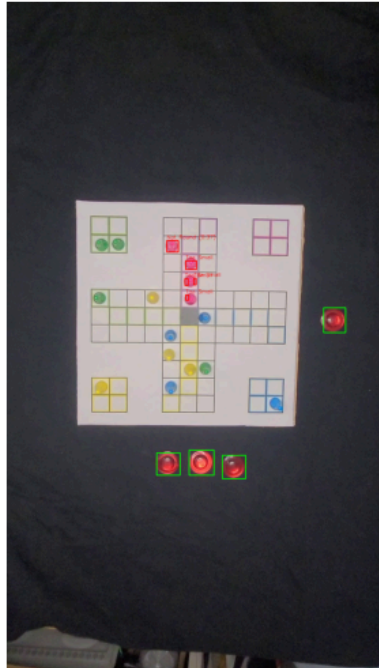
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Part 1: Red Color Segmentation (HSV)

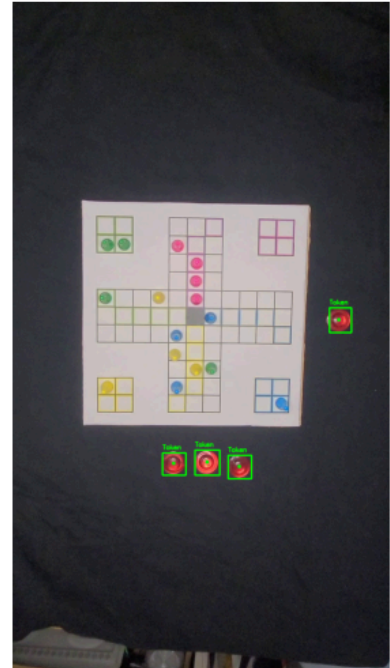


Part 2: Candidate Filtering & Final Tracking

5. Candidate Filtering
(Red=Rejected, Green=Accepted)



6. Final Active Trackers



4. Effectiveness

In the end we implemented detection of 5 events for each level of dataset difficulty:

- Dice roll - Reading the value on the dice
- Pawn entering the board - and bonus turn (rolling 6)
- Capture opponent/Coin collection - A pawn lands on an opponent's pawn, sending it back to home base, capturing player receives a coin token
- Game start - All pawns are positioned in their respective home bases
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In addition, we track the:

- Where the players' pawns are currently placed on board (Home/Track/Finish)
- Tracking the dice/Rolling

4.1 Easy

Our solution was very effective on the easy difficulty. There were no issues with tracking, only when a player's hand covered the board the pieces would disappear for a while but without any false detections.

4.2 Medium

The solution performed similarly for this difficulty. The lighting conditions didn't cause big issues for detecting the board and its pieces. The overall accuracy was almost the same as for the easy dataset.

4.3 Difficult

As for the highest difficulty the solution also performed really well in detecting pawns, board and tokens. However the hand movements covering the pieces and the angle, sometimes cause the pieces to be detected in the adjacent tiles or disappear more often than in lower difficulties. The biggest issue we saw with the dice value detection as the angle showed the side of the dice which caused some inaccuracies in the value detection.

5. Analysis and Conclusion

Throughout this project, we developed a computer vision pipeline designed to detect dice values and track game tokens on a Man Don't Get Angry board. While the system demonstrated great performance in identifying tokens and reading dice pips under controlled conditions, the development process highlighted critical challenges related to object segmentation.

A primary technical challenge was the accurate detection of the board grid. Initial attempts using the standard Hough Line Transform proved unreliable, the algorithm struggled to consistently identify grid lines amidst the visual noise of the pawns and varying lighting conditions, often resulting in fragmented, missing segments or non-existing lines. To overcome this, we shifted to a "projection peak" method. By summing pixel intensities along the x and y axes, we identified clear peaks corresponding to grid lines. This approach proved far more stable and deterministic than Hough lines, allowing us to reconstruct the grid structure reliably even when individual line segments were faint.

Similarly, tracking player pawns presented significant difficulties. Our initial approach relied solely on color intensity thresholding, which was highly vulnerable to environmental changes: shadows, hand movement, or slight shifts in lighting frequently caused false positives or lost tracking. We addressed this by implementing a multi-stage pipeline. The current system integrates HSV color space segmentation, strict geometric filtering based on circularity and area, and Euclidean distance tracking. This evolution from simple intensity checks to a comprehensive geometric and temporal tracking system was critical for maintaining consistent pawn identities across frames.

Despite these difficulties, the final implementation achieved significant accuracy through all recording. The integration of peak-based grid detection ensured the game board was correctly mapped, while the enhanced tracking logic allowed for the reliable monitoring of game pieces, the project demonstrated the potential of combining computer vision methods to analyze complex, dynamic recordings effectively.

In []: